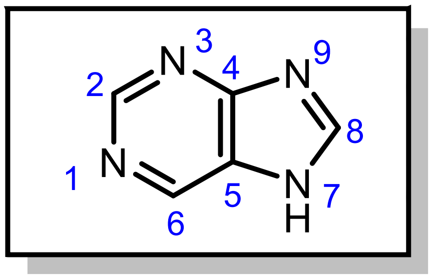
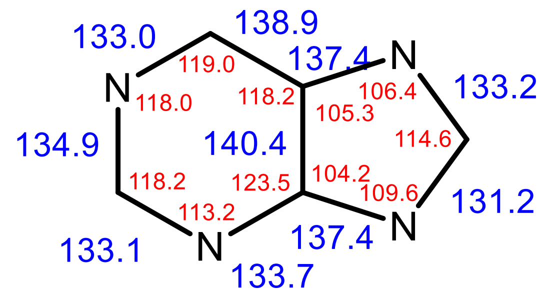
5.4. Anellierte Sechsring-Heterocyclen mit mehreren Heteroatomen

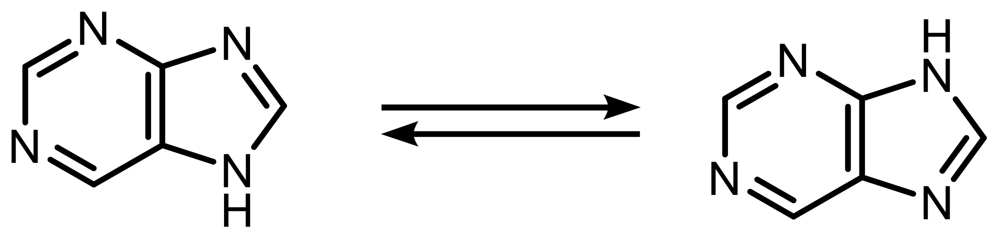
5.4.1. Purine



Bindungslängen [pm]
Bindungswinkel [°]



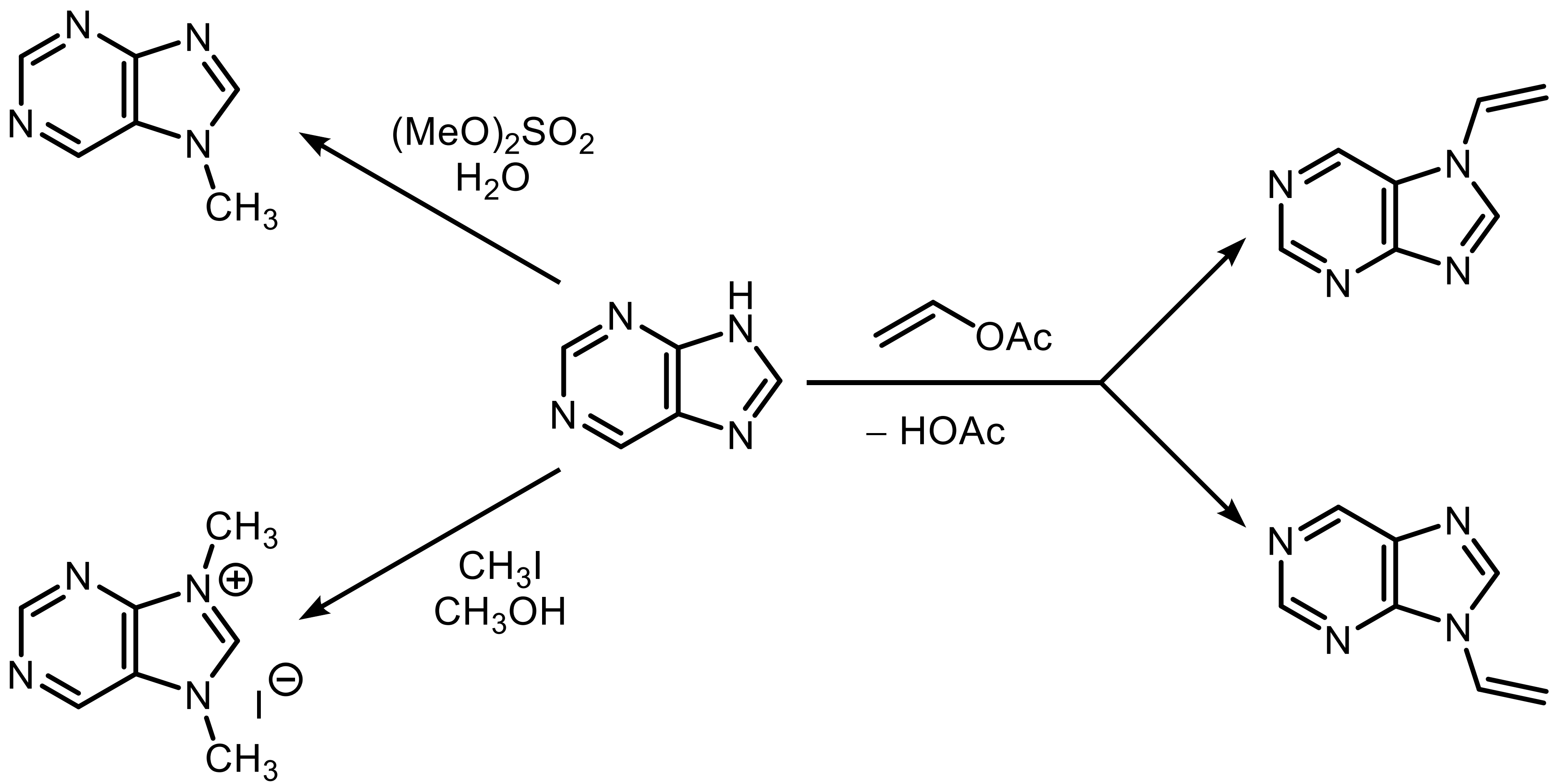
Tautomerie



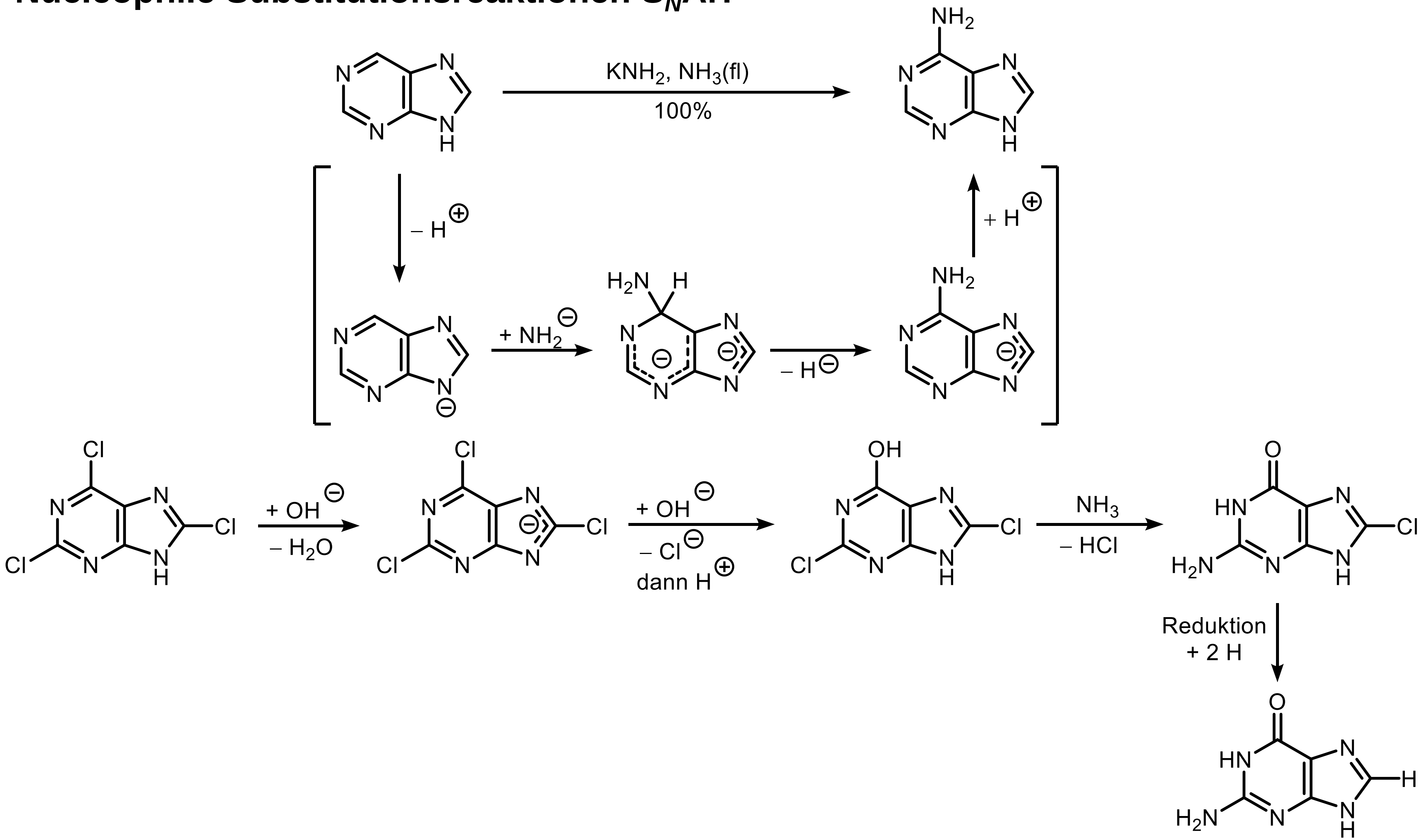
7*H*-Purin 9*H*-Purin

Reaktionen

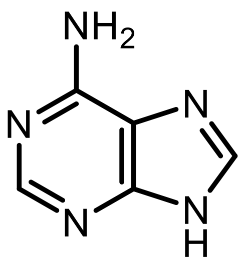
Mit Elektrophilen: an den N-Atomen



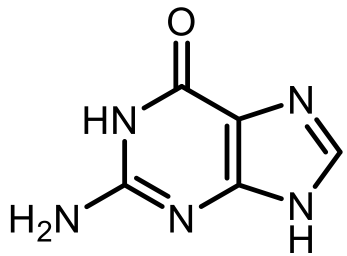
Nucleophile Substitutionsreaktionen S_NAr:



5.4.1. Purine
Vorkommen

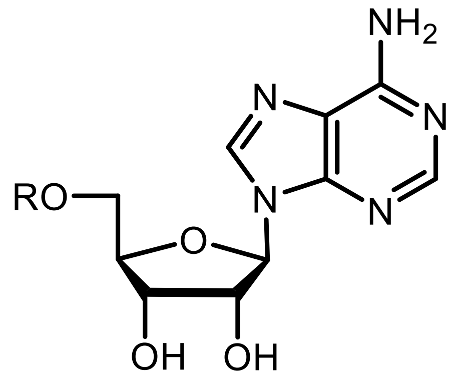


Adenin



Guanin

("Purinbasen" der Nucleinsäuren)

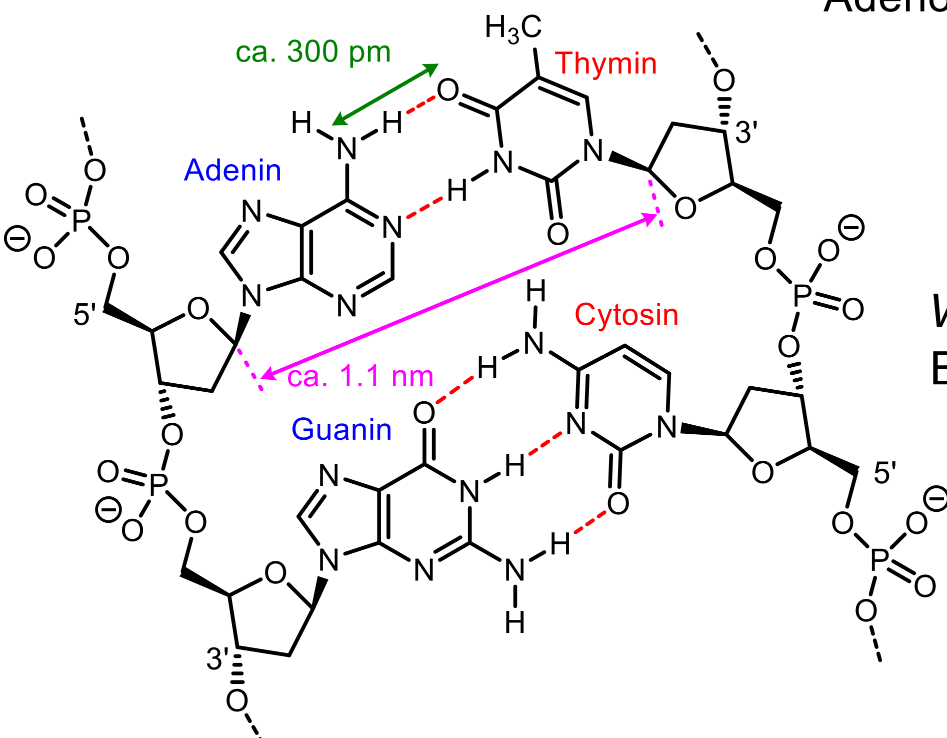


Adenosin (R = H)

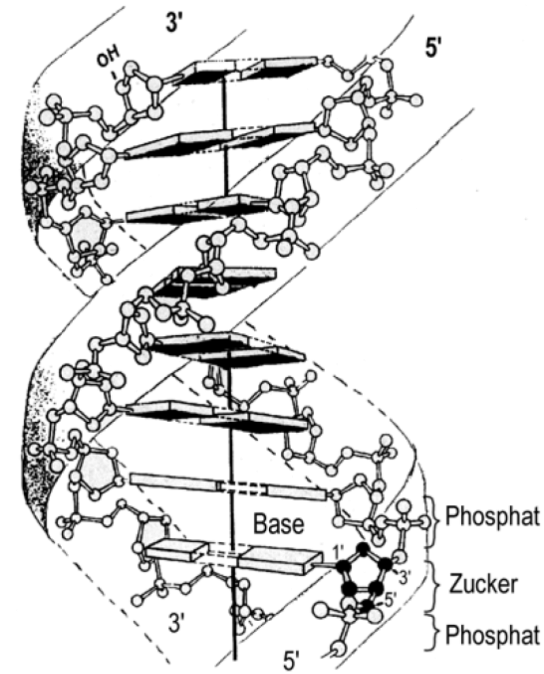
Adenosinmonophosphat (AMP)
(R = P(O)$(OH)_2$)

Adenosindiphosphat (ADP)
(R = P(O)(OH)OP(O)$(OH)_2$)

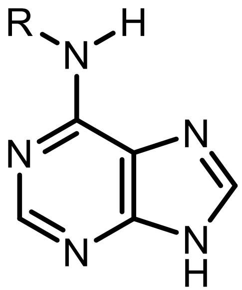
Adenosintriphosphat (ATP)
(R = P(O)(OH)OP(O)(OH)OP(O)$(OH)_2$)

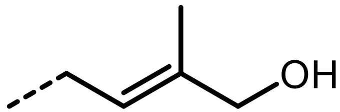


Watson-Crick-
Basenpaarung



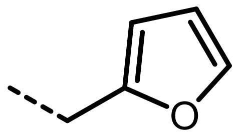
5.4.1. Purine
Wirkstoffe



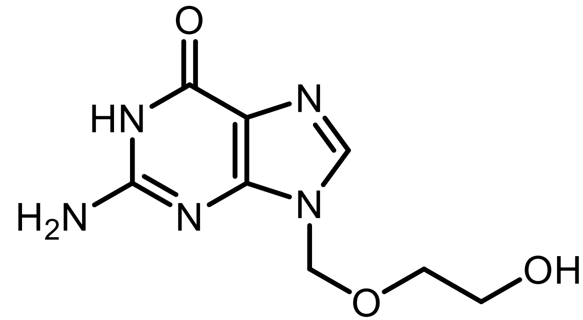
R = 

Zeatin (Cytokinin aus Mais)

R = CH_2Ph Verdan (in der Gemüsefrischhaltung)

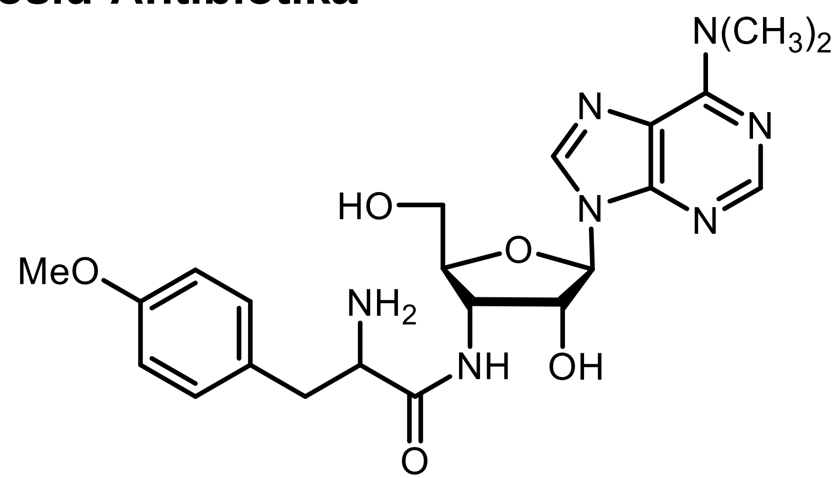
R = 

als "Anti-Aging"-Mittel für die Haut

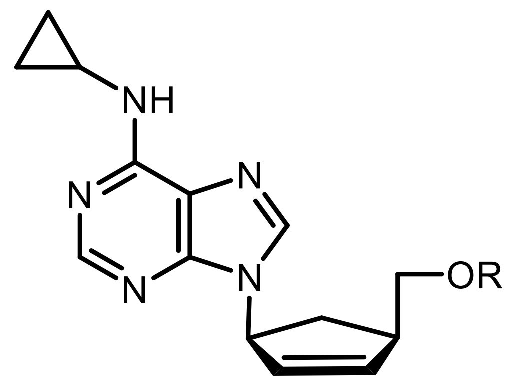


Acyclovir
(gegen Herpes simplex)

Nucleosid-Antibiotika

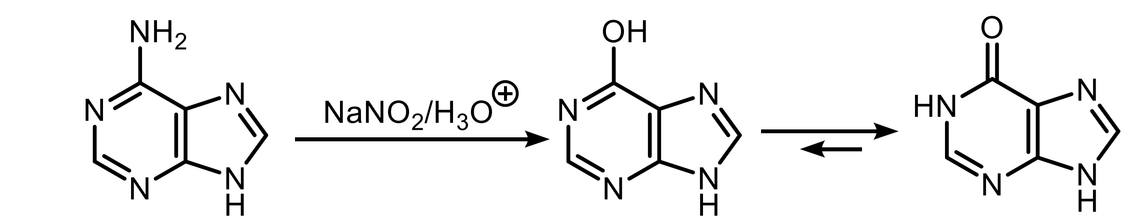


Puromycin
(Inhibitor der Proteinbiogenese in Bakterien und Säugetierzellen)



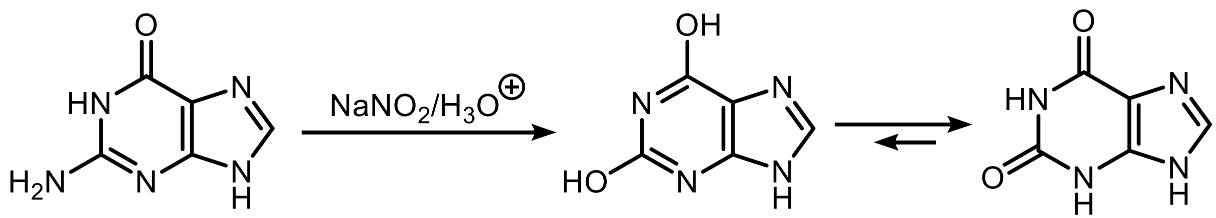
Abacavir
(potentes anti-HIV-Mittel)

5.4.1. Purine
Hydroxypurine



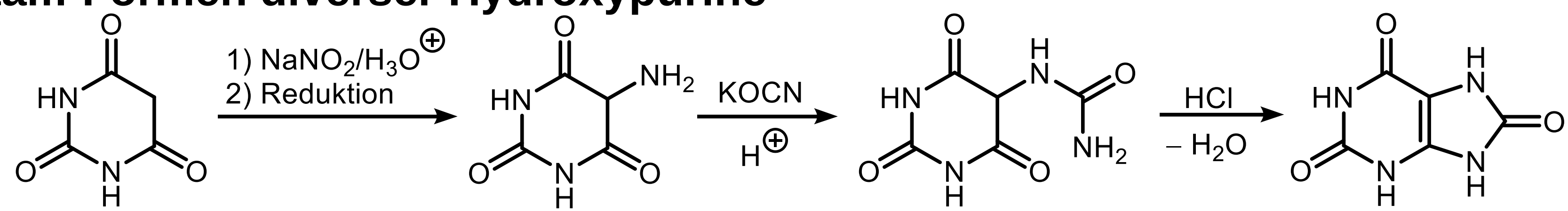
Lactim-Form Lactam-Form

Hypoxanthin

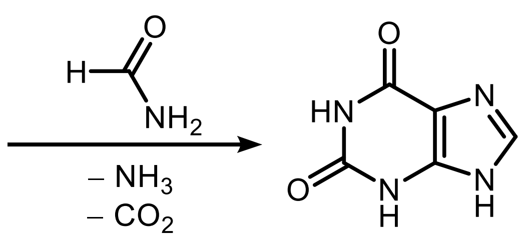


Xanthin

Lactam-Formen diverser Hydroxypurine

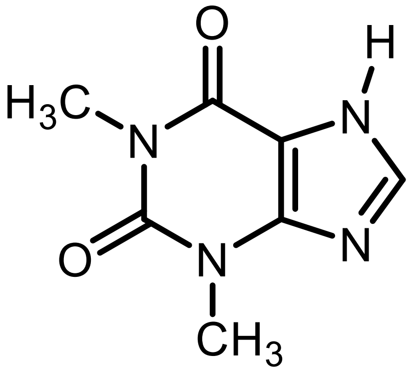


Barbitursäure Harnsäure

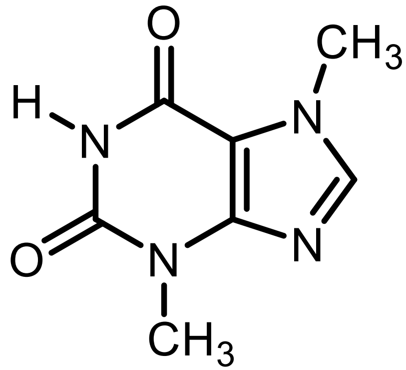


Xanthin

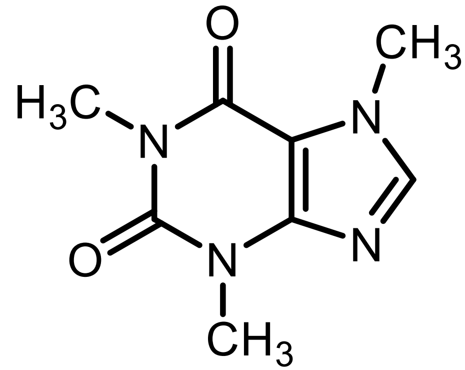
5.4.1. Purine
Purinalkaloide



Theophyllin



Theobromin



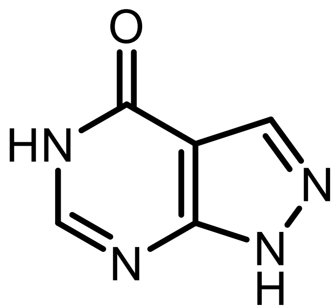
Coffein

Coffein (= 1,3,7-Trimethylxanthin)

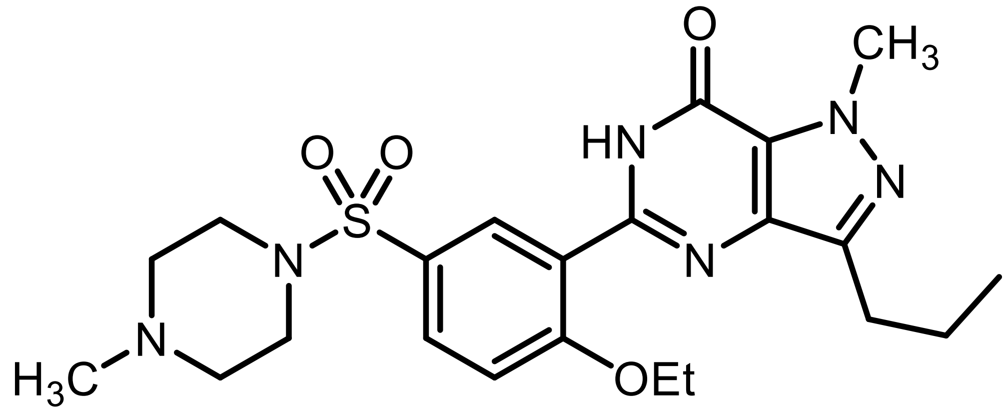
Isolierung: Runge, 1820.

Coffein hemmt die Phosphodiesterase, die cyclisches AMP (Adenosinmonophosphat) in AMP umwandelt, besitzt somit eine erregende Wirkung auf das ZNS. Herztätigkeit, Stoffwechsel und Atmung werden angeregt. Blutdruck und Körpertemperatur steigen, die Blutgefäße im Gehirn erweitern sich, die der Eingeweide verengen sich, weshalb von Müdigkeit verscheucht wird, sich die Stimmung hebt und die Leistungsfähigkeit gesteigert wird. Coffein ist auch bekannt für seine diuretische (harntreibende) Wirkung.

Isostere des Purins: z. B Pyrazolopyrimidine

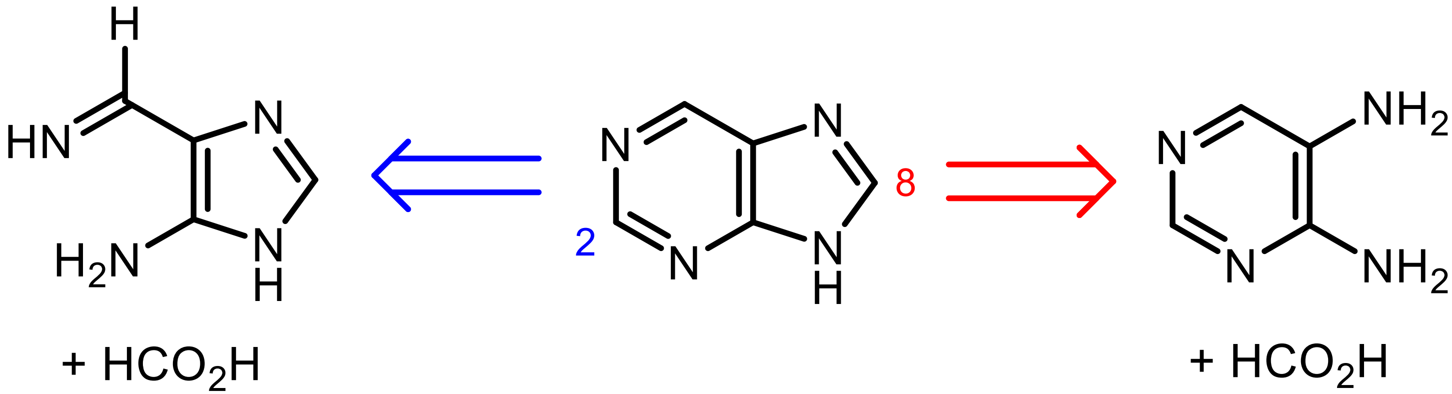


Allopurinol
(Behandlung von Gicht)

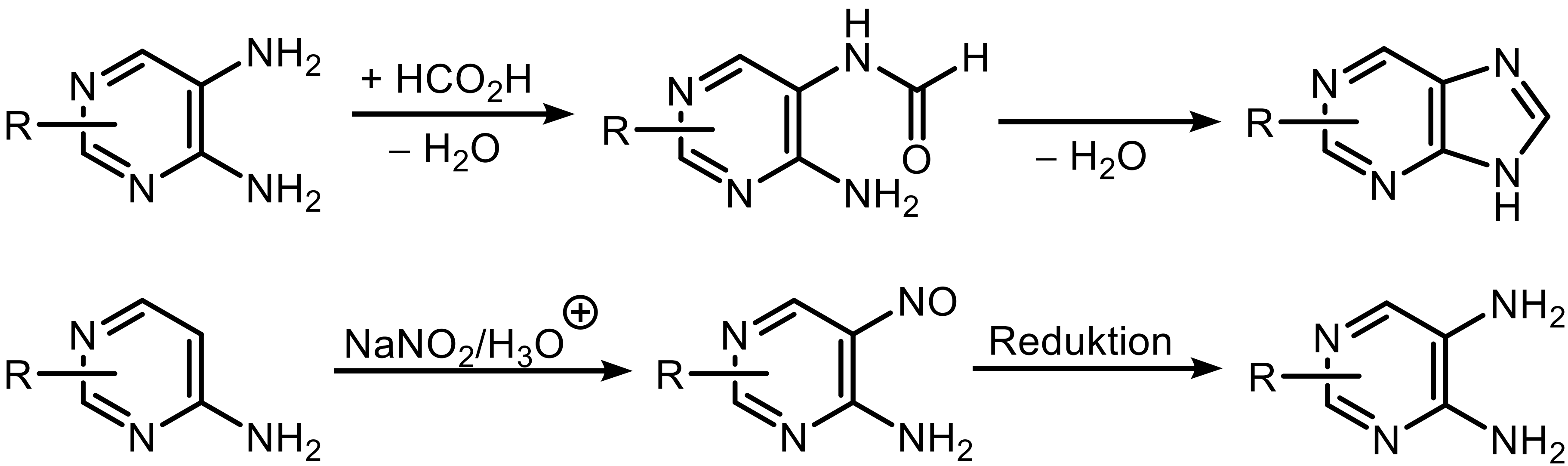


Sildenafil
(Inhibitor der Typ V cGMP-Phosphodiesterase)

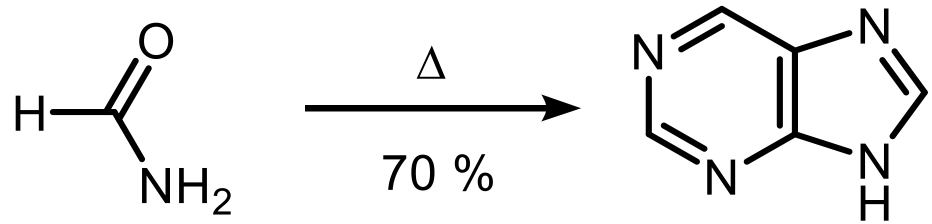
Retrosynthese



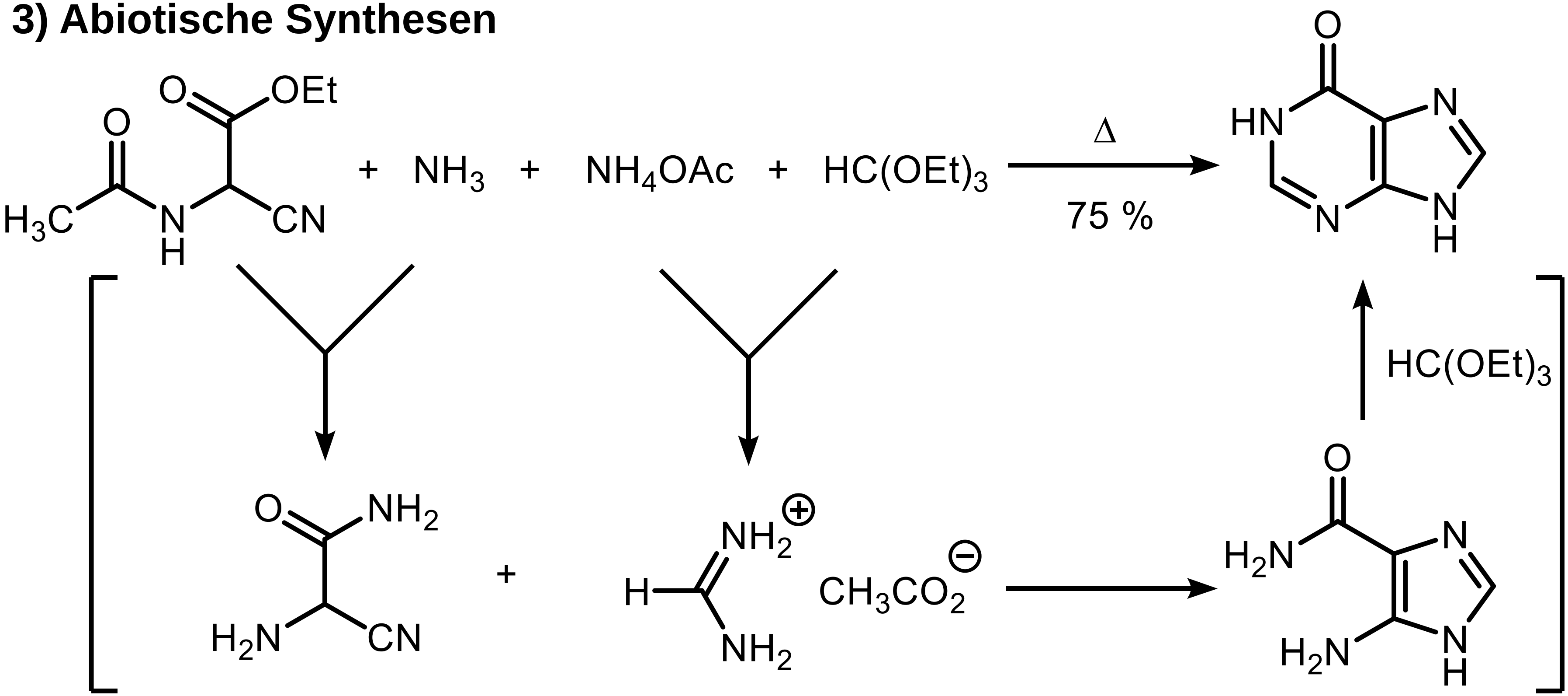
1) Traube-Synthese (1910)



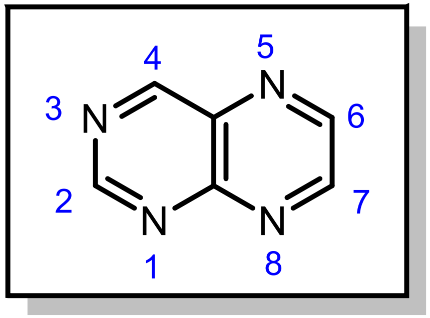
2) aus einfachen acyclischen Bausteinen



3) Abiotische Synthesen

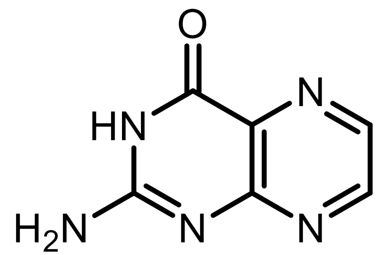


5.4.2. Pteridine

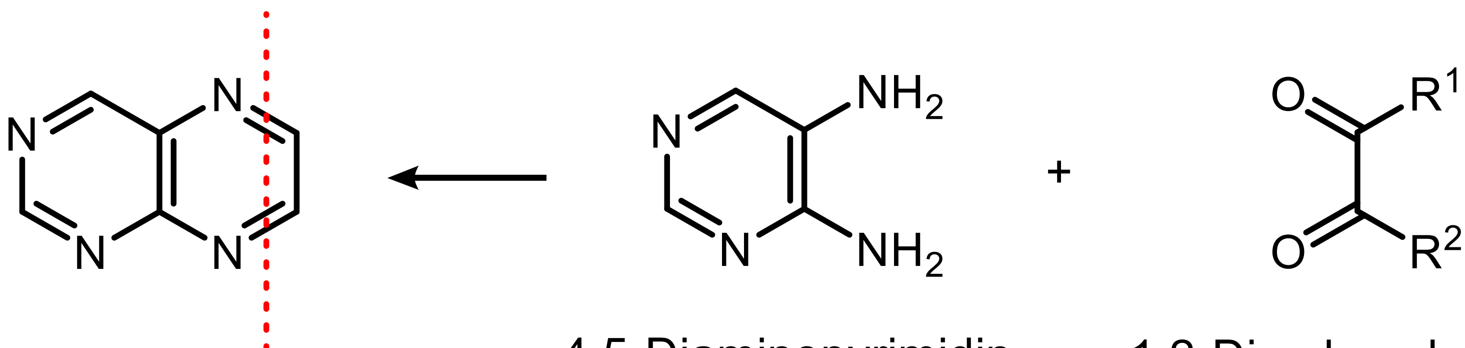


Pteridin (gelb)

Formal: Pyrimidin + Pyrazin (IUPAC: Pyrazino[2,3-*d*]pyrimidin
Extrem elektronenarm (Tieffeldverschiebung H, C im NMR)
Vorkommen: Farbstoffe der Schmetterlinge (Pterine), Coenzyme



Pterin

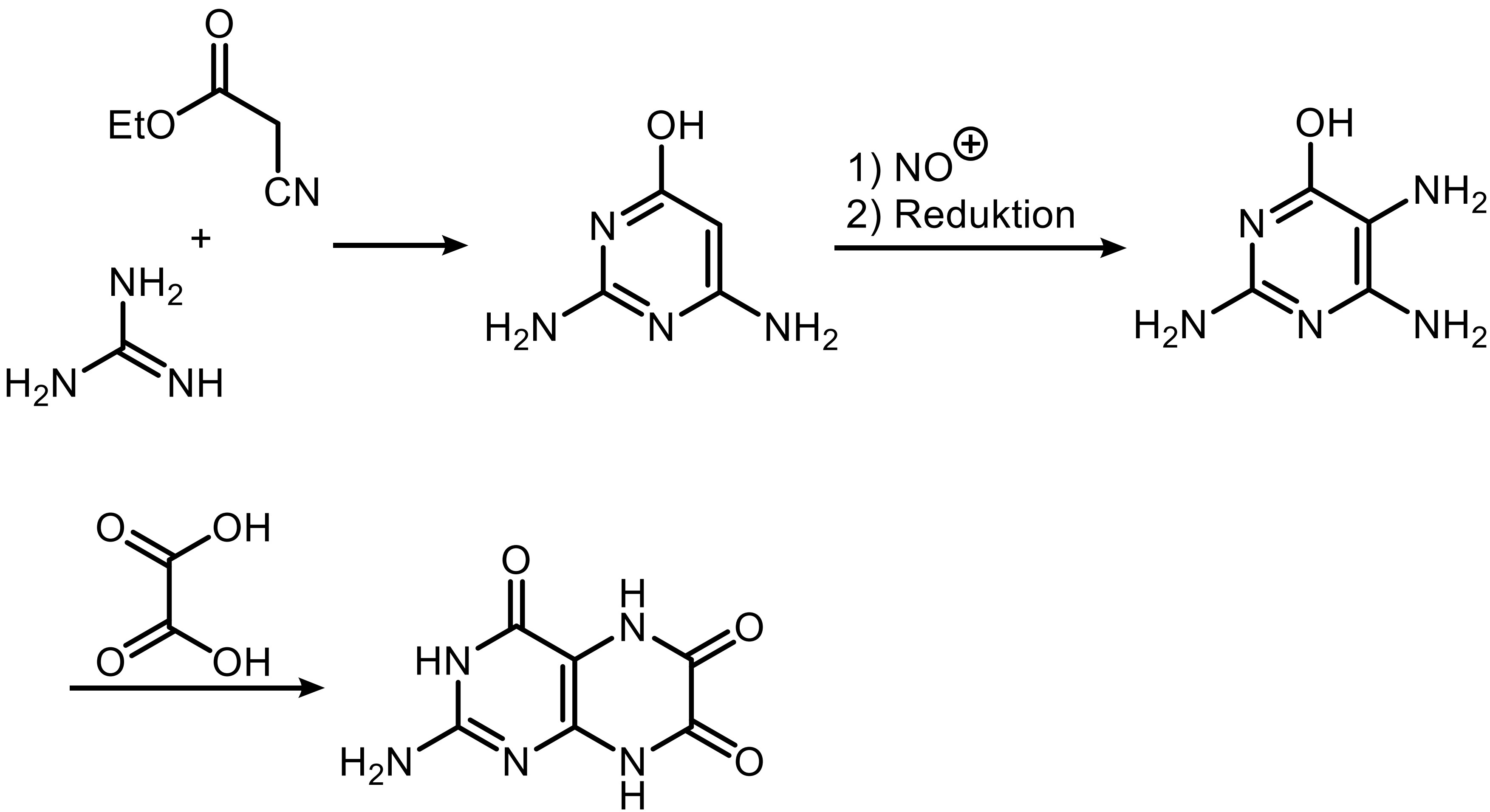


4,5-Diaminopyrimidin

1,2-Dicarbonylverbindung

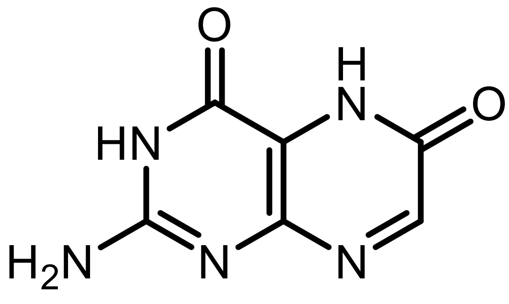
5.4.2. Pteridine
Synthesen

Leukopterin (weißer Farbstoff des Kohlweißlings, 1926 aus 200.000 (!) Schmetterlingen isoliert)

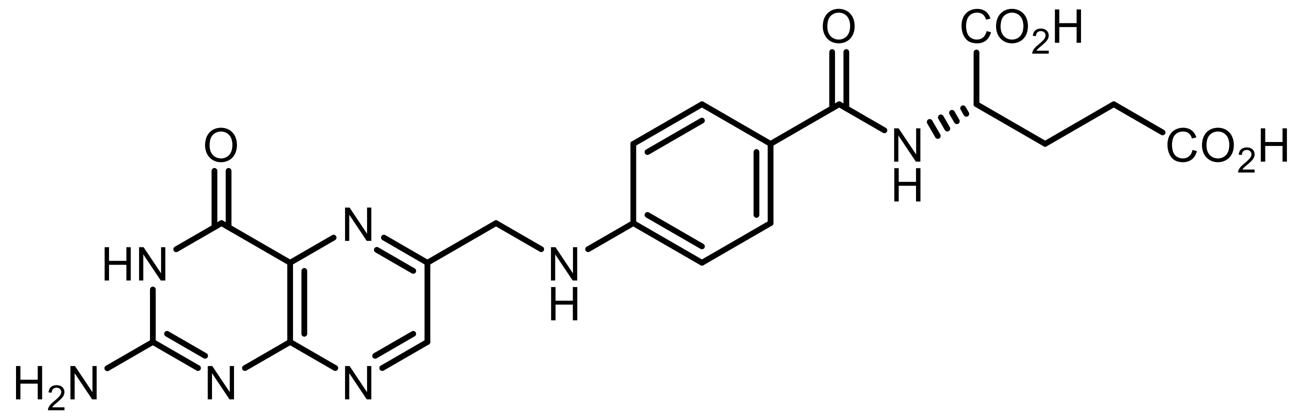


Leucopterin
(weiß)

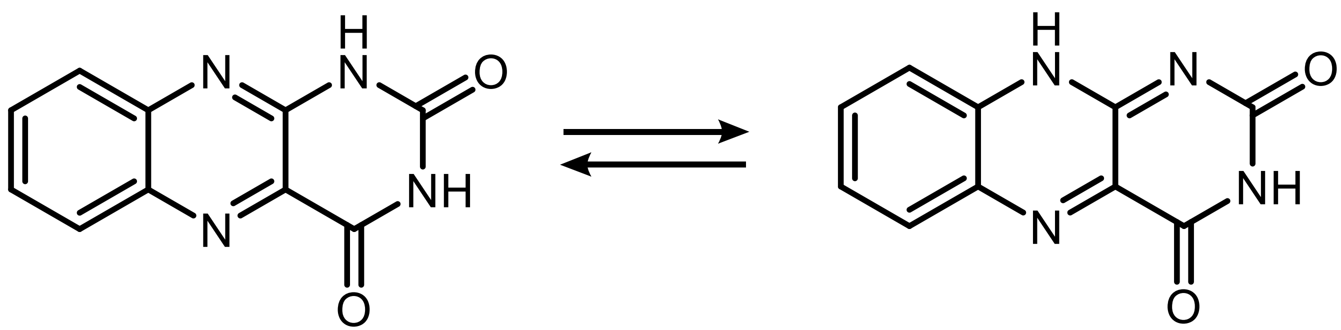
5.4.2. Pteridine
Vorkommen



Xanthopterin
(gelb, Zitronenfalter)



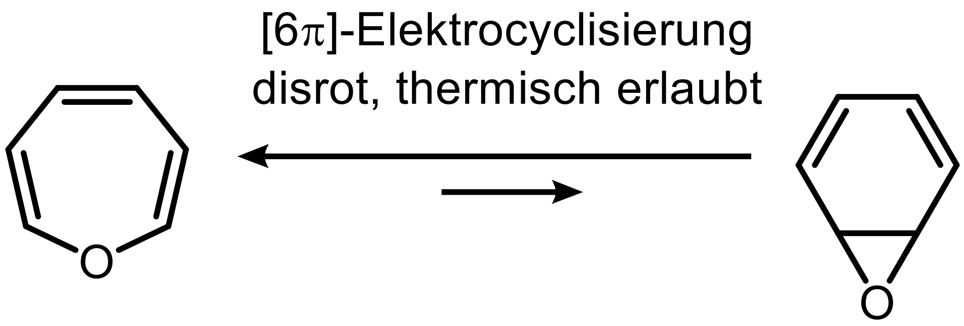
Folsäure
(orange, Vitamin der B-Gruppe)



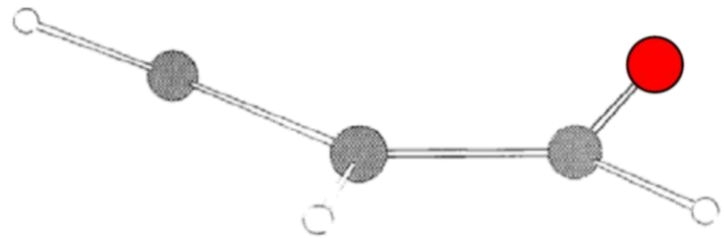
Alloxazin

Isalloxazin (Flavin)

6. Ausgewählte Siebenring-Heterocyclen
6.1. Oxepin

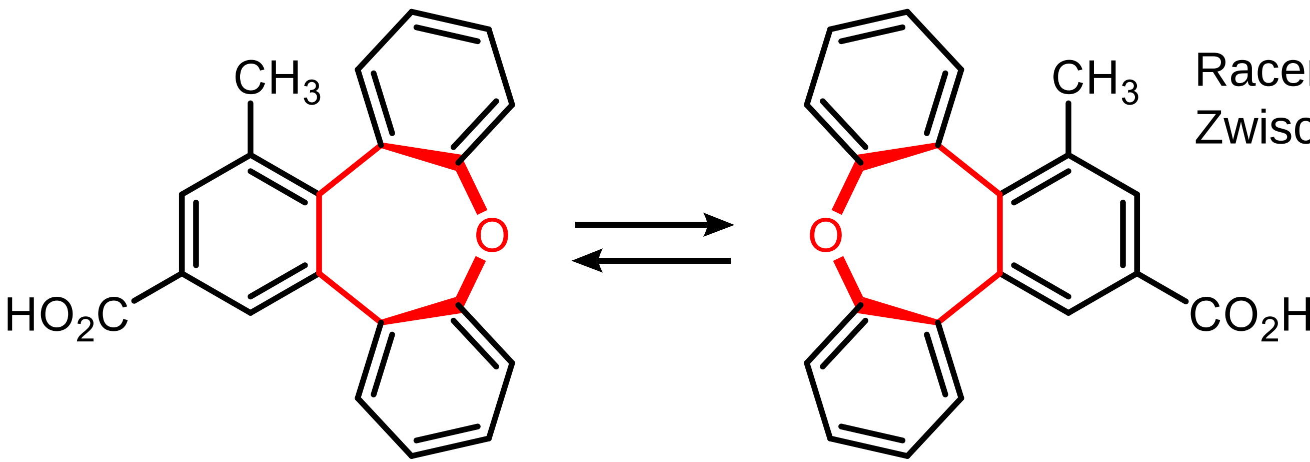


Oxepin   Benzoloxi



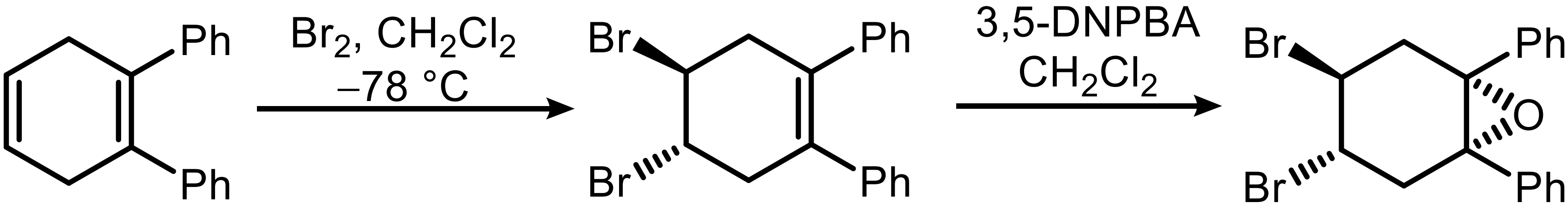
Toyota et al.
J. Phys. Chem. A **2003**, *107*, 2749.

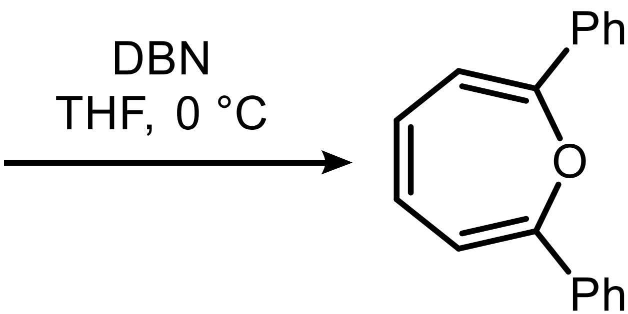
8π, lokalisierte C=C-Bindungen (Bindungslängenalternanz), gewinkelt (Boot-C_S-Geometrie) ⇒ Nicht-Aromat

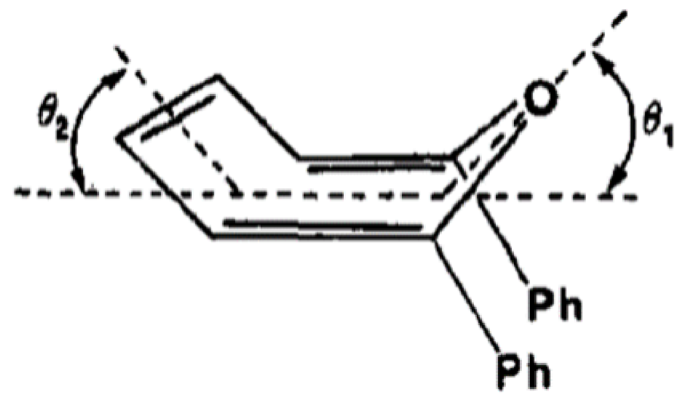


Racemisierung über Benzoloxid-Zwischenstufe ($\Delta G^{\neq}$ = 87 kJ/mol)

Synthese von 2,7-Diphenyloxepin

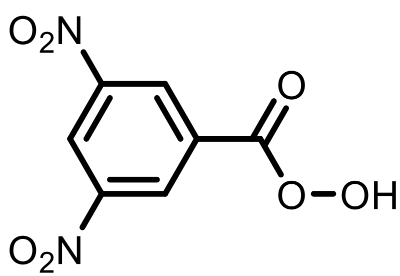




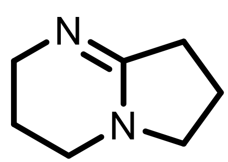


Röntgenstrukturanalyse

3,5-DNPBA

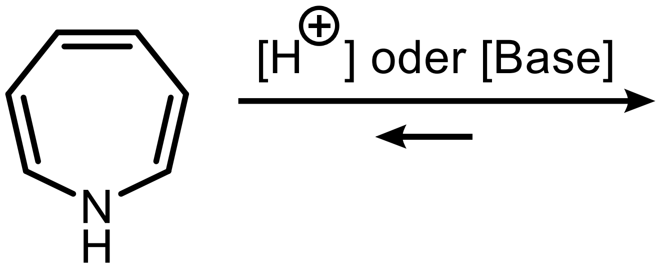


DBN

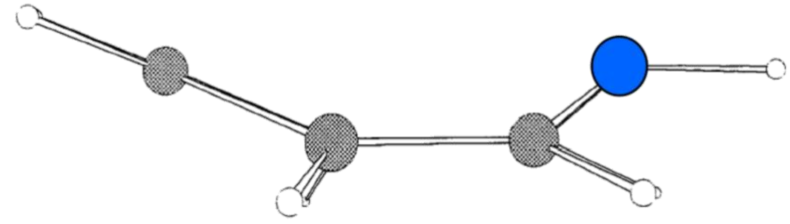


Boyd et al., *J. Org. Chem.* **1986**, 2784.

6.2. Azepin



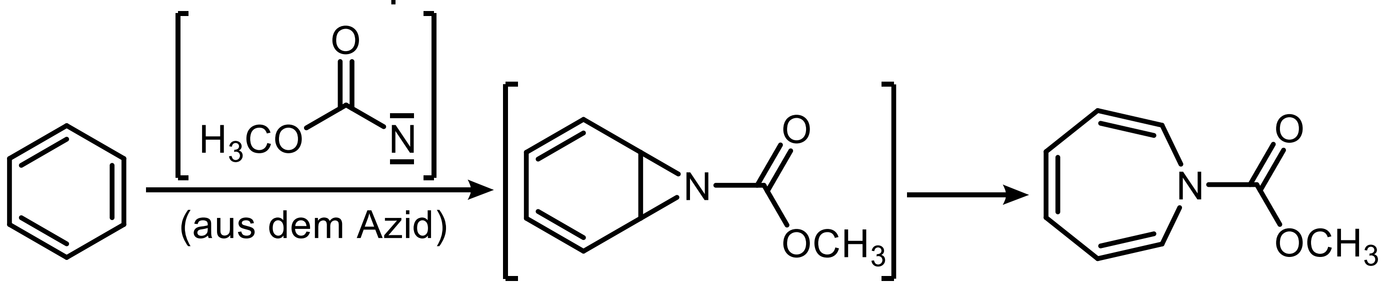
1*H*-Azepin (instabil) 3*H*-Azepin

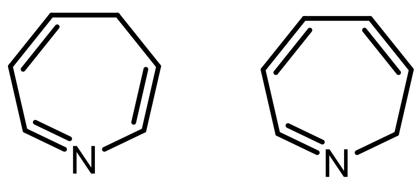


Toyota et al.
J. Phys. Chem. A **2003**, *107*, 2749.

8π, lokalisierte C=C-Bindungen (Bindungslängenalternanz), gewinkelt (Boot-C_S-Geometrie) ⇒ Nicht-Aromat

N-Amid-substituierte 1H-Azepine sind stabil

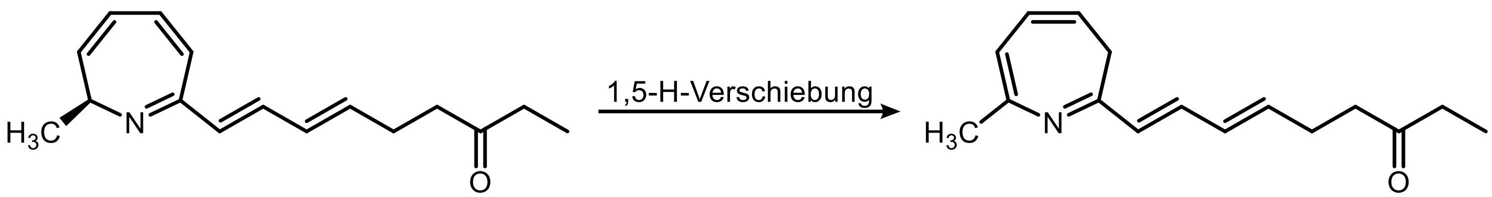




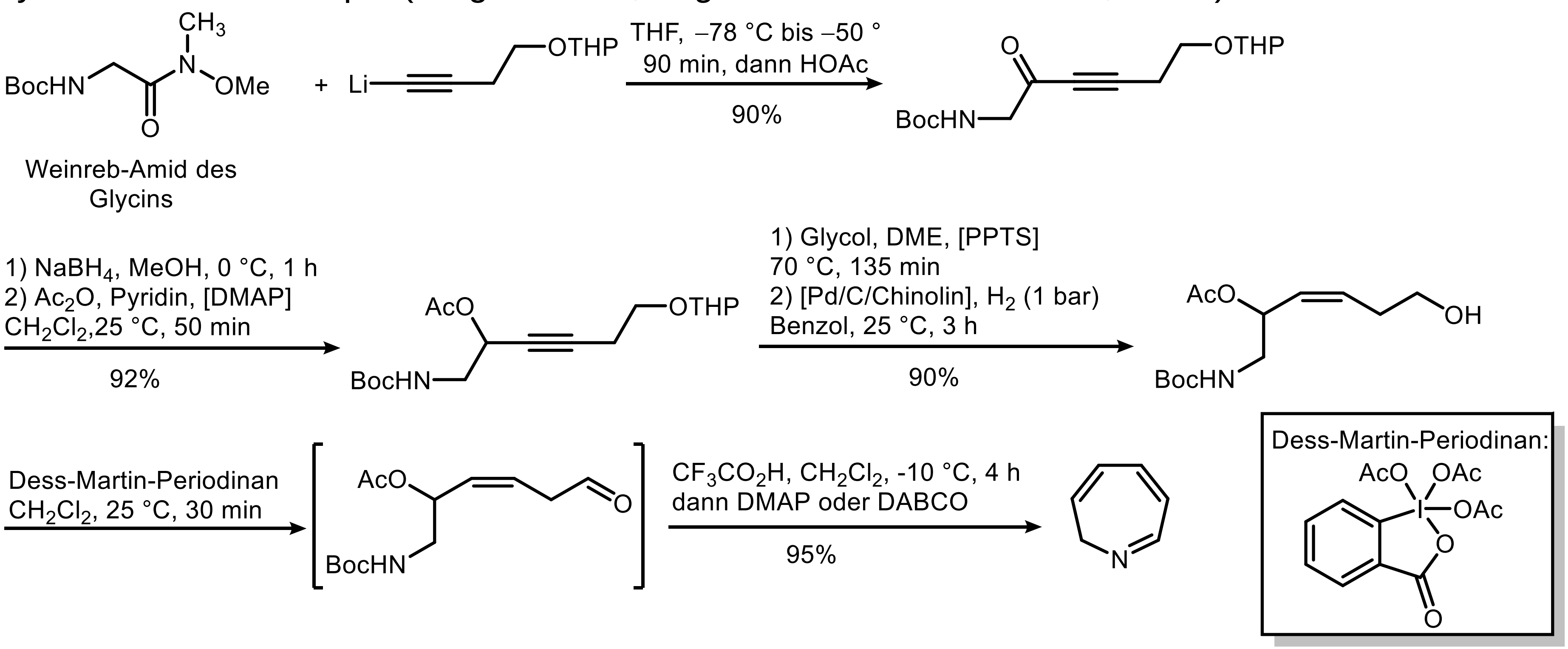
4*H*-Azepin 2*H*-Azepin

6.2. Azepin

2*H*-Azepin als Teilstruktur des Naturstoffs Chalciporon aus dem Pfefferröhrling (Chalciporus piperatus)

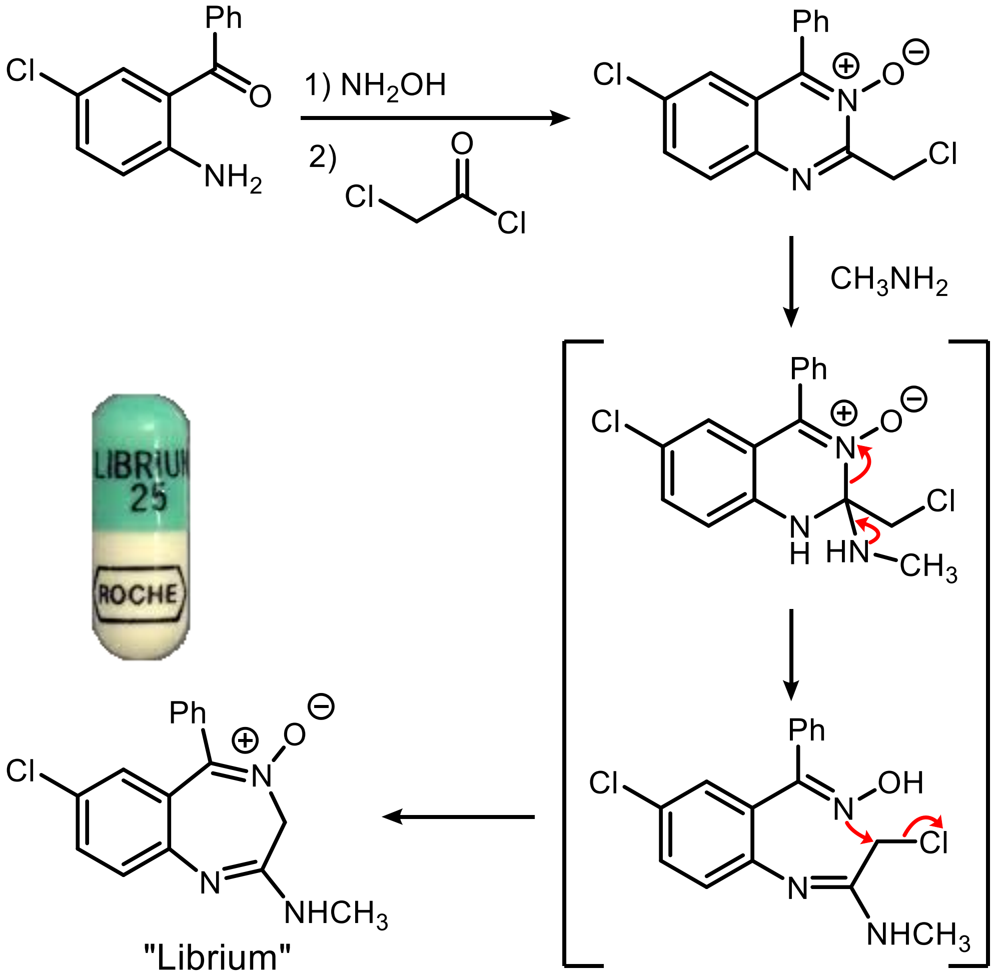


Synthese von 2*H*-Azepin (Steglich et al., *Angew. Chem. Int. Ed.* **1995**, 1469.)

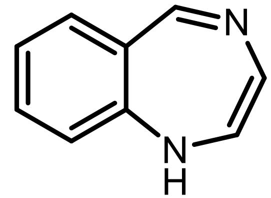


6.3. Diazepin

Leo Sternbach, Roche (1950er)



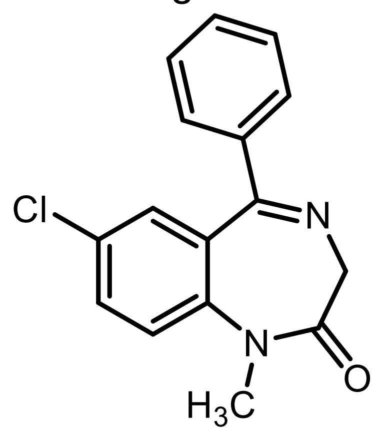
"Librium"
(1,4-Diazepinoxid)



Benzo-1,4-diazepin

Tranquilizer

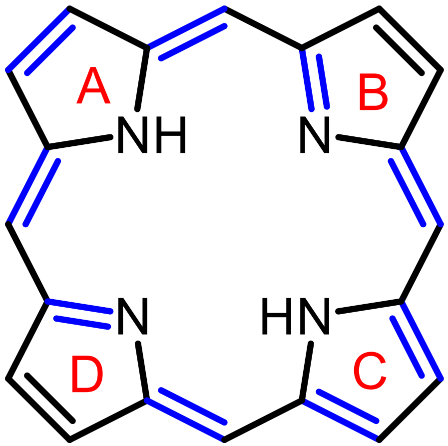
Benzodiazepine wirken im Allgemeinen: anxiolytisch (angstlösend), antikonvulsiv (krampflösend), muskelrelaxierend (muskelentspannend), sedativ (beruhigend und schlaffördernd), amnestisch (Erinnerung für die Zeit der Wirkdauer fehlt) und leicht stimmungsaufhellend.



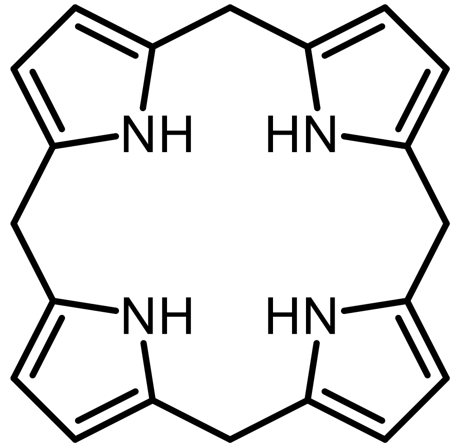
Diazepam
(Valium, 10 x aktiver)

Benzodiazepine gelten als die Medikamente mit der höchsten Missbrauchsrate in Deutschland.

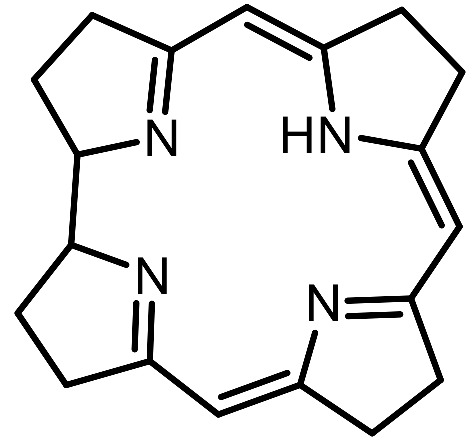
7. Porphyrin – ein makrocyclischer Heterocyclus



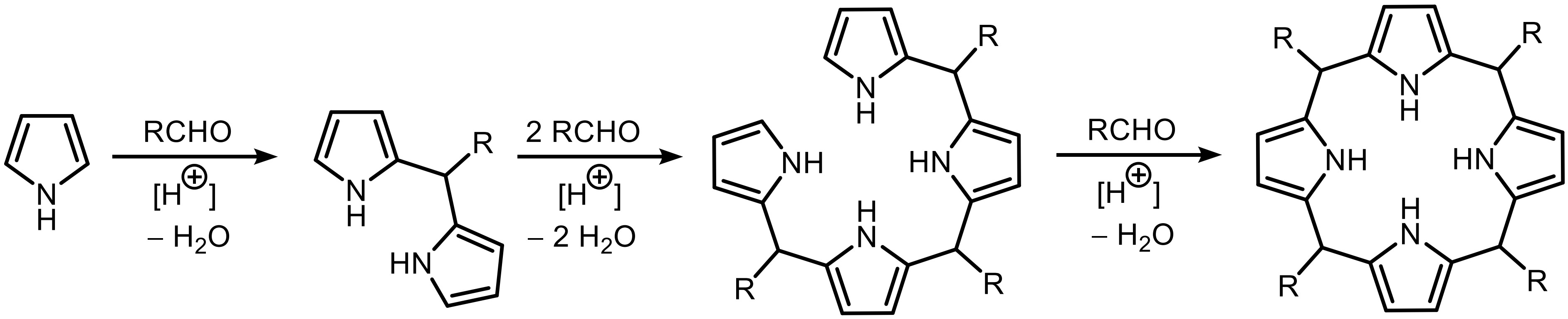
Porphyrin
(18π-System)

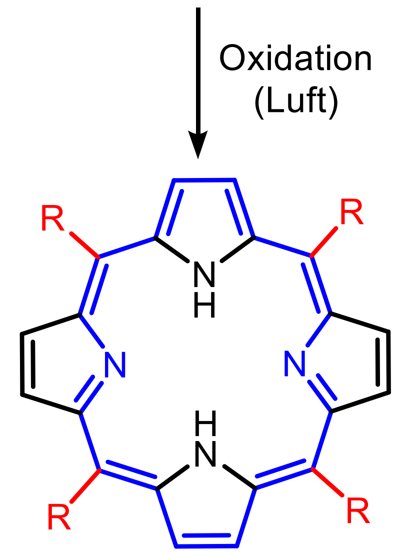


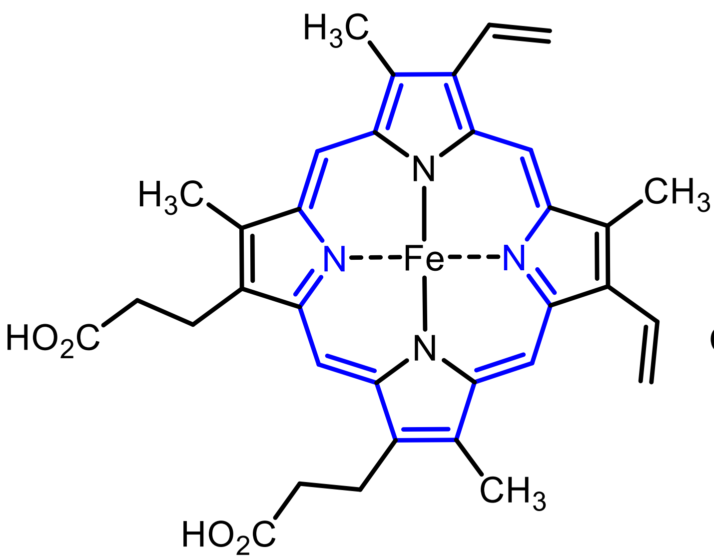
Porphyrinogen
(Hexahydroporphyrin)



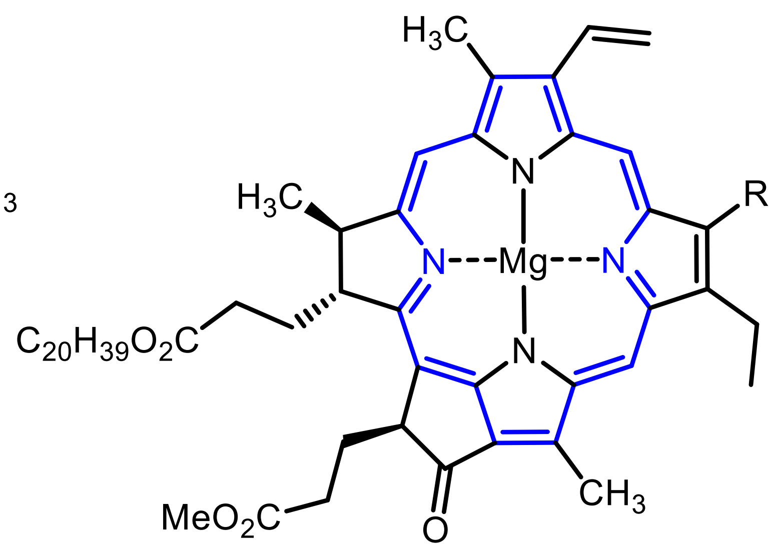
Corrin







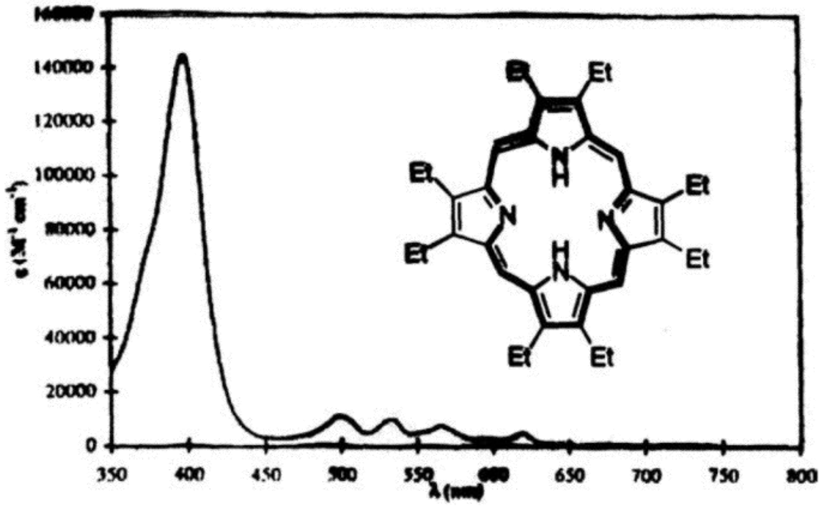
Häm

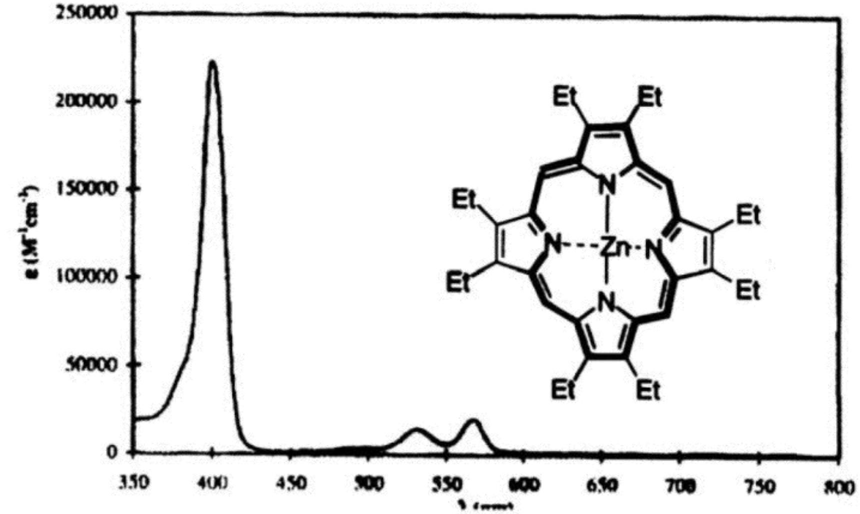


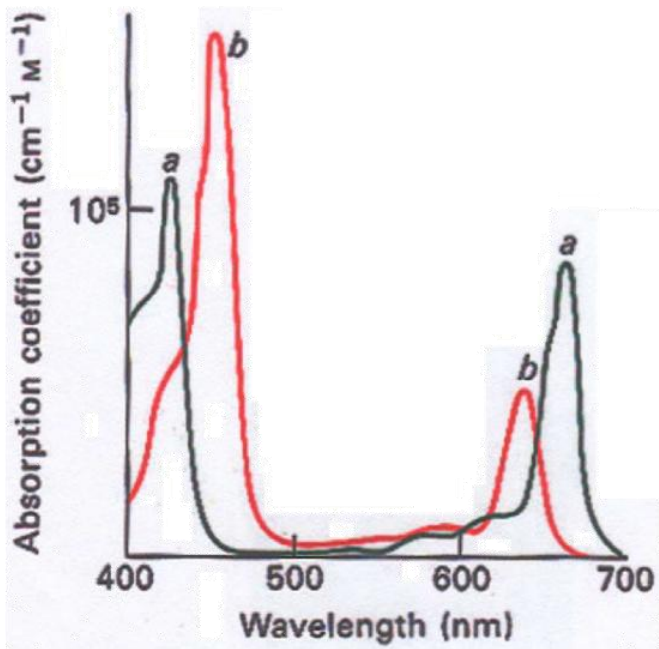
Chlorophyll a (R = CH_3)
Chlorophyll b (R = CHO)

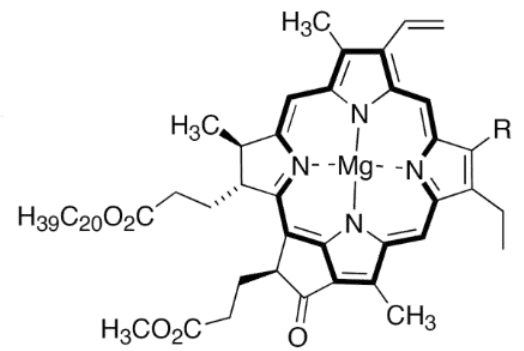
Porphyrin
(18π-System)
Substituenten R in
meso-Positionen

7. Porphyrin



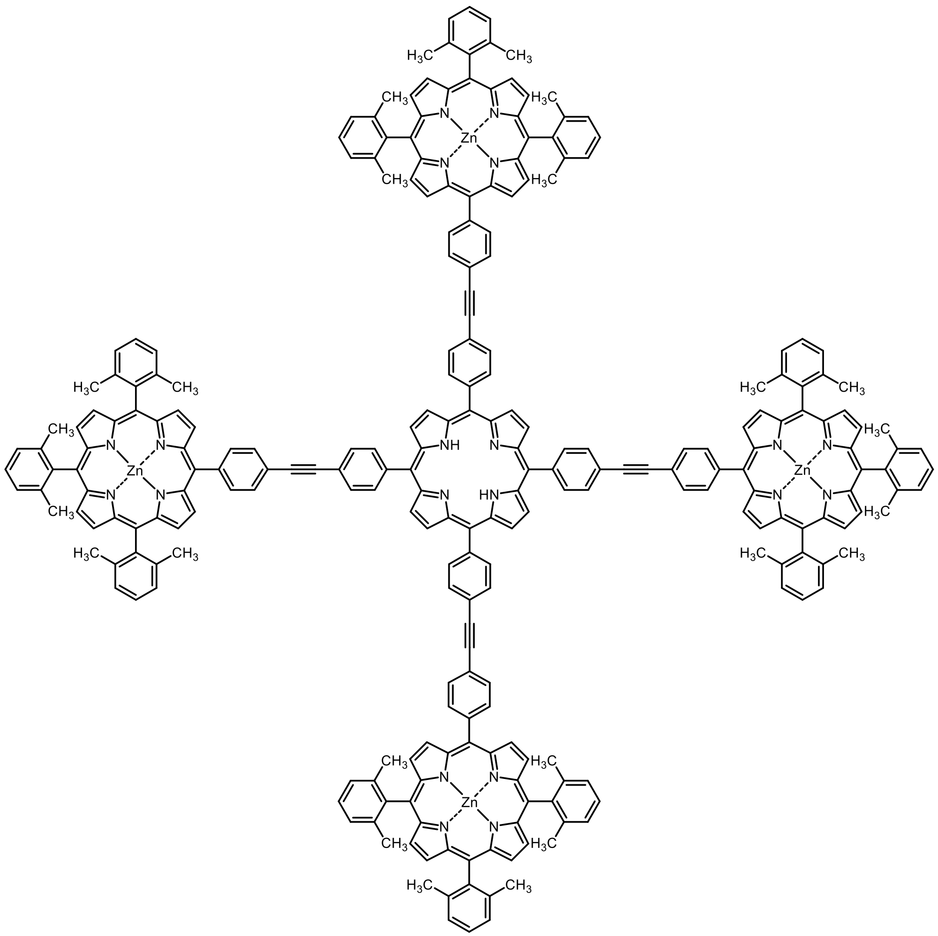


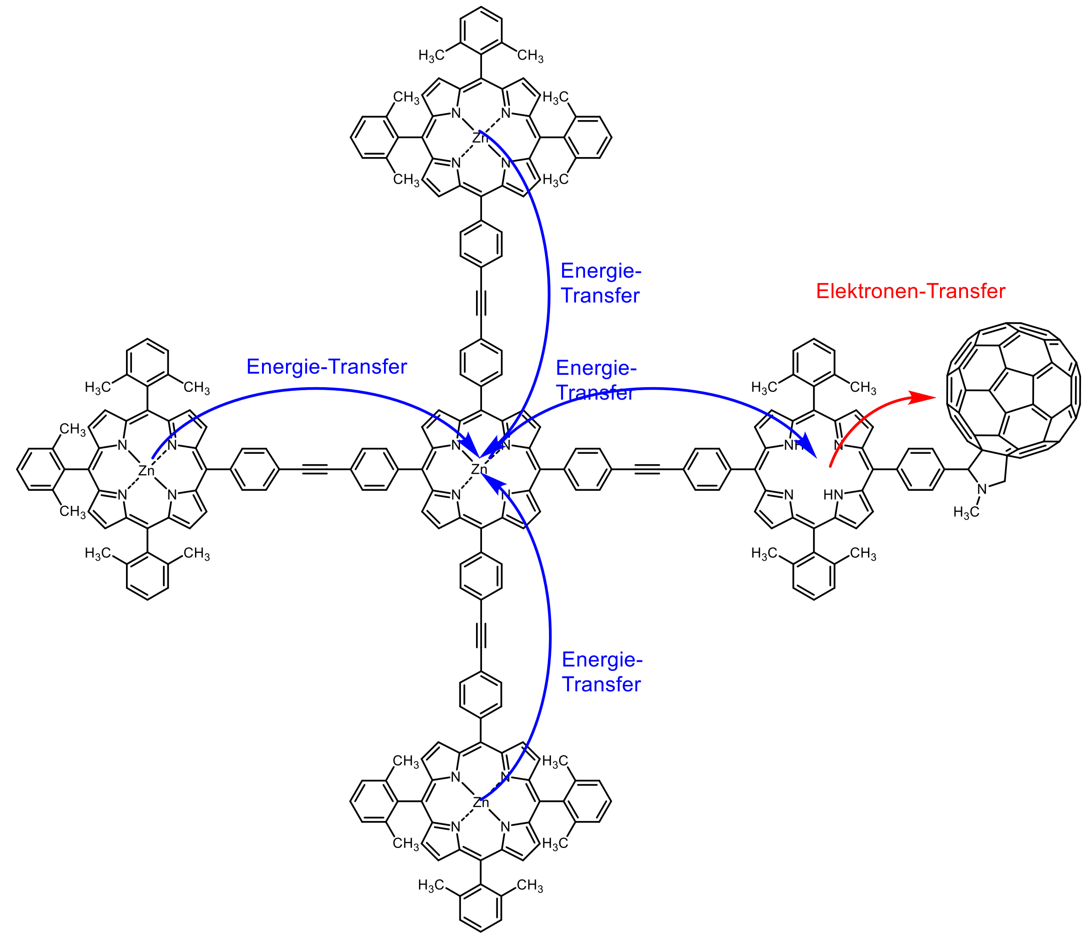




Chlorophyll a: R = CH_3
Chlorophyll b: R = CHO

7. Porphyrin





Effizienter Energietransfer vom Zn-Porphyrin zum inneren Porphyrin (95-99%)
J. Am. Chem. Soc. **1993**, *115*, 7519.

Lebensdauer des ladungsgetrennten Zustands: 240 ns
J. Phys. Chem. A **2002**, *106*, 2036.